

Cosmology: Problems

1. Horizon Problem

Consider a flat universe filled with only matter and radiation. The Friedmann equation gives us the evolution of the Hubble rate

$$H^2 = H_0^2 \left(\Omega_m a^{-3} + \Omega_r a^{-4} \right), \quad (1)$$

where $\Omega_r = a_{\text{eq}} \Omega_m$, with $a_{\text{eq}} \approx 3400^{-1}$ the scale factor at matter-radiation equality.

- (a) Show that the conformal time as a function of the scale factor then is

$$\eta = \frac{2}{\sqrt{\Omega_m}} H_0^{-1} \left(\sqrt{a + a_{\text{eq}}} - \sqrt{a_{\text{eq}}} \right). \quad (2)$$

Compute the conformal times today ($a_0 = 1$) and at recombination ($a_{\text{rec}} \approx 1100^{-1}$).

- (b) By comparing the comoving distance to the surface of last-scattering to the size of the comoving horizon at recombination, compute the angular size θ_h of a causally connected region on the last-scattering surface.
- (c) How would the result change if you included the effects of dark energy? (For a quantitative estimate, the relevant integrals have to be computed numerically.)
- (d) Explain how your analysis relates to features of the CMB power spectrum.
- (e) Please go to the website: <https://chrisnorth.github.io/planckapps/Simulator/>. You will find a simulation that shows how the CMB power spectrum responds to changes in the baryon density (Ω_b), the dark matter density (Ω_c) and the dark energy density (Ω_Λ).

How do the peak positions change when you increase Ω_Λ (at fixed Ω_m)? Does this agree with your expectation from part (c)?

2. Massless Field in De Sitter

Consider a massless scalar field ϕ in four-dimensional de Sitter space.

1. We first explore how the form of n -point correlation functions $\langle \phi(\mathbf{x}_1) \cdots \phi(\mathbf{x}_n) \rangle$ is constrained by the symmetries of the spacetime.
- (a) Invariance under translations, $\mathbf{x} \rightarrow \mathbf{x} + \mathbf{b}$, where $\mathbf{b}$ is a constant vector, implies that the correlators must satisfy $\langle \phi(\mathbf{x}_1) \cdots \phi(\mathbf{x}_n) \rangle = \langle \phi(\mathbf{x}_1 + \mathbf{b}) \cdots \phi(\mathbf{x}_n + \mathbf{b}) \rangle$. Using this fact, prove that the momentum-space correlators must obey

$$\langle \phi_{\mathbf{k}_1} \cdots \phi_{\mathbf{k}_n} \rangle = F_n(\mathbf{k}_1, \dots, \mathbf{k}_n) \delta(\mathbf{k}_1 + \cdots + \mathbf{k}_n), \quad (1)$$

where $\delta(\cdots)$ is the Dirac delta function.

- (b) The de Sitter metric in conformal time is invariant under dilatations, $(\eta, \mathbf{x}) \rightarrow \lambda(\eta, \mathbf{x})$, where λ is a constant. Since the late-time correlators of a massless scalar field do not

depend on time, they must satisfy $\langle \phi(\mathbf{x}_1) \cdots \phi(\mathbf{x}_n) \rangle = \langle \phi(\lambda \mathbf{x}_1) \cdots \phi(\lambda \mathbf{x}_n) \rangle$. Show that the functions F_n in (1) must then scale as

$$F_n(\lambda \mathbf{k}_1, \dots, \lambda \mathbf{k}_n) = \frac{1}{\lambda^{3(n-1)}} F_n(\mathbf{k}_1, \dots, \mathbf{k}_n). \quad (2)$$

What does this imply for the power-law scaling of the two-point function?

2. Next, we actually compute the two-point function of the quantum field in the vacuum state. The mode expansion of the field operator is

$$\hat{\phi}_{\mathbf{k}}(\eta) = f_k(\eta) \hat{a}_{\mathbf{k}} + f_k^*(\eta) \hat{a}_{-\mathbf{k}}^\dagger, \quad (5)$$

where $[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{k}'}^\dagger] = (2\pi)^3 \delta(\mathbf{k} - \mathbf{k}')$. The Bunch–Davies mode function $f_k(\eta)$ was derived in Problem 2.

- (a) Derive the two-point function $\langle 0 | \hat{\phi}_{\mathbf{k}}(\eta) \hat{\phi}_{\mathbf{k}'}(\eta) | 0 \rangle$ in the limit $\eta \rightarrow 0$, where $|0\rangle$ is the vacuum defined by $\hat{a}_{\mathbf{k}}|0\rangle = 0$.
- (b) Show that the real space two-point function $\langle \phi(\mathbf{x}) \phi(\mathbf{0}) \rangle$ is IR divergent.

3. Flat-Space Wavefunction

Consider a massless scalar in flat space with a cubic interaction:

$$S = \int d^4x \left(-\frac{1}{2} (\partial\Phi)^2 - \frac{\lambda}{3!} \Phi^3 \right). \quad (1)$$

We will study wavefunction coefficients, evaluated at a fixed time $t_* \equiv 0$.

- (a) Show that the three- and four-point wavefunction coefficients are

$$\psi_3 = \frac{\lambda}{K}, \quad (2)$$

$$\psi_4 = \frac{\lambda^2}{EE_L E_R} + t/u\text{-channels}, \quad (3)$$

where

$$\begin{aligned} K &\equiv k_1 + k_2 + k_3, \\ E &\equiv k_1 + k_2 + k_3 + k_4, \quad E_L \equiv k_{12} + q, \quad E_R \equiv k_{34} + q. \end{aligned} \quad (4)$$

- (b) With some effort, you could also compute the five-point wavefunction coefficient:

$$\begin{aligned} \psi_5 &= \text{Diagram: A horizontal purple line with five points labeled } k_1, k_2, k_3, k_4, k_5 \text{ from left to right. Below the line, there are two vertices. The first vertex is connected to } k_1, k_2, \text{ and } k_3. \text{ The second vertex is connected to } k_3, k_4, \text{ and } k_5. \text{ A horizontal line segment connects the two vertices, labeled } q \text{ below the first vertex and } q' \text{ below the second vertex.} \\ &= \frac{\lambda^3}{EE_L E_M E_R} \left[\frac{1}{k_{123} + q'} + \frac{1}{k_{345} + q} \right], \end{aligned} \quad (5)$$

where $E_M \equiv k_3 + k_I + k'_I$. Do you see a pattern in the results above? If so, use it to write down the answer for the six-point wavefunction coefficient ψ_6 without any computation.

- (c) In a unitarity theory, the wavefunction coefficients satisfy interesting cutting rules. For the four-point wavefunction coefficient (in the s -channel), the cutting rule is

$$\tilde{\psi}_4 = -P(q) \left(\tilde{\psi}_3 \times \tilde{\psi}_3 \right), \quad (6)$$

where we have defined

$$\tilde{\psi}_4 \equiv \psi_4(\underline{\mathbf{k}}) + \psi_4^*(-\underline{\mathbf{k}}), \quad (7)$$

$$\tilde{\psi}_3 \equiv \psi_3(\underline{\mathbf{k}}, q) + \psi_3^*(-\underline{\mathbf{k}}, q). \quad (8)$$

In the complex conjugated wavefunction coefficients, we have analytically continued the external energies $\underline{\mathbf{k}} \rightarrow -\underline{\mathbf{k}}$. Verify by explicit substitution that the relation (6) holds for the results in part (a).